

Negotiation Protocol between Alice and Bob

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1 Objective

Alice wants to sell her car, Bob is interested in buying it. Alice asks some quote. Bob can accept the bargain, refuse it or make a counteroffer. Alice can accept or make a counteroffer and so on, Until either the bargain is accepted or refused.

Bob can:

1. accept the proposal;
2. refuse the proposal;
3. send a counteroffer.

If Bob sends a counteroffer, Alice can:

1. accept it;
2. send another counteroffer.

2 Diagrammatic Notation

Table 1 defines the visual notation used in the model.

Element	Visual notation	Meaning
Alice	Light-blue region	States and actions belonging to Alice
Bob	Light-green region	States and actions belonging to Bob
Start	Circle	Beginning of the negotiation
Decision	Diamond	Evaluation of a received proposal
Quote or counteroffer	Blue arrow	A price proposal is sent to the other actor
Accept	Green arrow	The current proposal is accepted
Refuse	Red arrow	The negotiation is refused
Successful termination	Double green circle	Agreement reached
Failed termination	Double red circle	Negotiation terminated without agreement

Table 1: Visual notation for the negotiation protocol.

The notation separates two concepts: **shapes represent states**, whereas **arrows represent messages or actions**. The colour of an arrow expresses the meaning of its message: blue denotes a proposal, green denotes acceptance, and red denotes refusal. The coloured regions identify the actor responsible for the enclosed states.

3 Negotiation Protocol

Alice chooses an initial price p and sends it to Bob:

$$\text{Alice} \xrightarrow{\text{Quote}(p)} \text{Bob}.$$

Bob evaluates Alice's proposal and has three alternatives:

Accept

Bob accepts the proposed price, and the negotiation terminates successfully:

$$\text{Bob} \xrightarrow{\text{Accept}} \text{Agreement}.$$

Refuse

Bob refuses the proposal, and the negotiation terminates without an agreement:

$$\text{Bob} \xrightarrow{\text{Refuse}} \text{Rejected}.$$

Counteroffer

Bob proposes a different price p' and transfers the turn to Alice:

$$\text{Bob} \xrightarrow{\text{Counteroffer}(p')} \text{Alice}.$$

After receiving Bob's counteroffer, Alice has two alternatives. She may accept it, terminating the negotiation successfully, or propose another price p'' :

$$\text{Alice} \xrightarrow{\text{Accept}} \text{Agreement}, \quad \text{Alice} \xrightarrow{\text{Counteroffer}(p'')} \text{Bob}.$$

In the second case, control returns to Bob and the negotiation continues.

4 Diagram

Figure 1 shows the complete protocol. Alice and Bob are placed in separate coloured regions, while the two terminal outcomes are shared by the whole process.

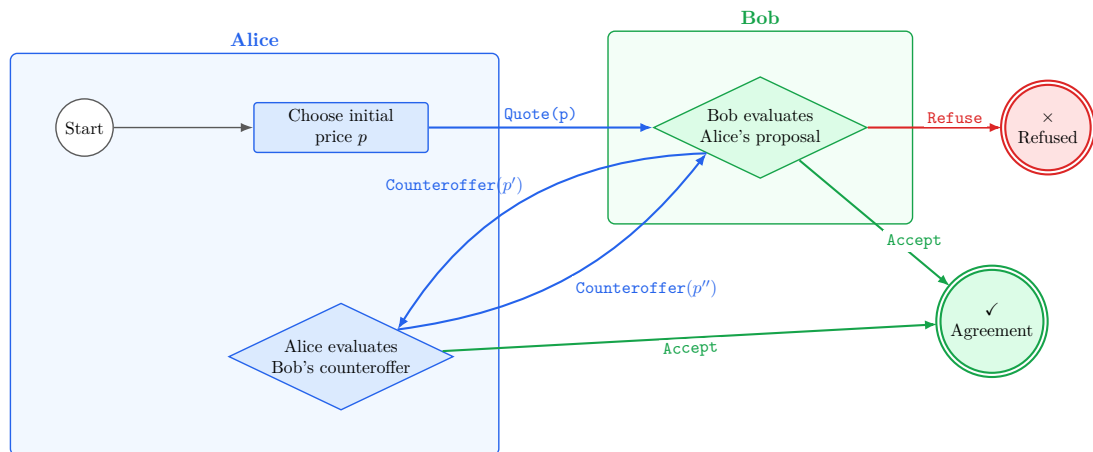


Figure 1: Diagram of the negotiation protocol.

5 Properties of the Model

5.1 Absence of dead states

Every non-final state has at least one outgoing transition. In particular,

$$\text{Out}(B) = \{\text{Accept}, \text{Refuse}, \text{Counteroffer}\}, \quad \text{Out}(A) = \{\text{Accept}, \text{Counteroffer}\}.$$

Therefore, the protocol contains no intermediate state from which no action is possible.

5.2 Terminal states

The protocol has two terminal outcomes. *Agreement* represents successful termination after Alice or Bob accepts the current proposal. *Refused* represents unsuccessful termination after Bob refuses Alice's proposal. Neither terminal state has outgoing transitions.

5.3 Negotiation loop and termination

Counteroffers create a cycle between the two decision states. Consequently, the protocol **does not guarantee termination**: Alice and Bob may, in principle, exchange counteroffers indefinitely. The negotiation terminates only if Alice or Bob eventually sends **Accept**, or Bob sends **Refuse**. Thus,

$$\text{Accept} \vee \text{Refuse} \implies \text{termination},$$

but the infinite behaviour Counteroffer^n remains possible.